L2.3 How Derivatives Affect the Shape of a Graph *Recall: L2.2 The Mean Value Theorem*

Name ________________________________ **Solo – 9 minutes**

1. Verify that the function satisfies the three hypotheses of Rolle's Theorem on the given interval. Then find all numbers c that satisfy the conclusion of Rolle's Theorem.

$$f(x) = x + \frac{1}{x} \qquad \text{on } \left[\tfrac{1}{2},\, 2\right]$$

Check the hypotheses one at a time, and only then use the conclusion. Say why any candidate you find is kept or rejected.

2. Find the number c that satisfies the conclusion of the Mean Value Theorem.

$$f(x) = \sqrt{x} \qquad \text{on } [0, 4]$$

3. Suppose $f'(x) > 0$ for every x in an interval. In one sentence, what does that tell you about the graph of f on that interval? And if $f'(x) < 0$?